

Alan Jese Rivas Muñoz

St Louis, MO | +1 3145463398 | alanr@wustl.edu

[LinkedIn](#) | [GitHub](#) | [Academic Portfolio](#)

EDUCATION

Washington University in St. Louis – St. Louis, MO

PhD Student in Biomedical Engineering, 1st year student, *McDonnell International Scholar - Abhinav Jha Lab*

- Developing novel tool, to perform accurate quantification of LowDose PET Images.
- Building a whole body SPECT simulation model from scratch.

Instituto Tecnológico de Estudios Superiores de Monterrey (ITESM) – Monterrey, Mexico

B.Sc. in Physics Engineering, GPA: 91.22 / 100 – *Expected December 2024*

- Excellence List & Academic Talent Scholarship, Top 1% Student award
- Relevant coursework: Optics, Quantum Mechanics, Process Modeling, Phase Unwrapping for Imaging Systems

RESEARCH EXPERIENCE

Research Intern – Neuroimaging & AI

Harvard Medical School / Boston Children's Hospital – FNNDSC, Kiho Im Lab | Aug 2023 – Jun 2024

- Developed a novel deep learning U-Net–based architecture for multi-tissue segmentation of fetal brain MRI, achieving Dice similarity scores up to 0.99.
- Led a student research team in transitioning the MRI processing pipeline to support high-resolution fetal brain imaging, implementing a novel tissue-preserving resampling workflow that reduced voxel size from 0.86 mm to 0.50 mm isotropic, enabling more precise structural analysis.
- Gained experience in project coordination, documentation systems, and clinical quality control.

Research Intern – Nuclear Imaging & Optimization

Washington University in St. Louis – Abhinav Jha Lab | Jul – Sep 2024

- Investigated quantitative SPECT reconstruction with Lu-177 using OSEM method, determining optimal hyperparameters for task based, detection and quantification.
- Designed a custom anthropomorphic phantom to improve diagnostic reliability for outlier patients.
- Used MATLAB, Python and SIMIND for dataset simulation and image reconstruction.

PROFESSIONAL EXPERIENCE

Operations Specialist

Scale AI (Remote, US-based) | Sep 2024 – Aug 2025

- Managed over 10 cross-functional data projects with a combined value exceeding \$10M USD, collaborating closely with clients and technical teams.
- Introduced new data query and annotation tools, improving operational throughput by 3×.
- Led 4 key delivery projects for high-volume labeled datasets, ensuring on-time, quality-assured outcomes.

ENTREPRENEURIAL EXPERIENCE

Co-Founder – El Rey del Paste (\$800,000 revenue in 2025)

Monterrey, Mexico | 2022 – Present

- Launched and operated a regional food delivery business, specializing in traditional pastries.
- Developed standardized production processes, quality control checks, and staff training systems.

AWARDS

- McDonnell International Scholar Fellowship, WashU in St Louis - Sept 2025
- Top 1% Student Award, ITESM – Sept 2024